

YouTube Trending Video Analytics

Global Content, Engagement & Sentiment Analysis

1. Introduction

This project analyzes YouTube trending video data across five countries India, the United States, Germany, Great Britain and Mexico to understand content popularity, audience engagement, regional differences, trending duration, and title/tag sentiment. The project combines data cleaning, exploratory analysis, natural language processing, SQL analysis and interactive dashboarding to turn raw multi-country trending data into actionable insights.

2. Abstract

The raw multi-country datasets were consolidated into a unified collection of 198,508 records spanning all five regions. Following the removal of 4,531 duplicate entries, the refined dataset comprised 193,977 unique observations. Data pre-processing involved resolving corrupted identifier fields, verifying numerical and chronological attributes, aligning category mappings across regions, and normalizing textual elements for natural language processing. Using a multilingual XLM-RoBERTa architecture, title and tag strings were analyzed to quantify sentiment across positive, neutral, and negative metrics. SQL queries were subsequently executed to evaluate category distribution, cross-regional variations, time-series dynamics, and trending lifespans, with key findings rendered in an interactive four-page Power BI dashboard.

3. Tools Used:

Tools	Purpose
Google Colab / Python / Pandas	Data loading, cleaning, validation and exploratory analysis
Transformers / XLM-RoBERTa	Multilingual sentiment classification of titles and tags
SQLite / SQL	Category, country, duration, time-series and sentiment analysis
Power BI Desktop	Interactive dashboard and data storytelling
Excel	Quick data checks where useful

4. Steps Involved

- **Data Collection and Integration:** Loaded the India, United States, Germany, Great Britain, and Mexico YouTube trending CSV files and combined them into one dataset.
- **Data Cleaning and Validation:** Removed exact duplicates, standardized dates, checked numeric engagement fields, validated publish/trending date consistency, handled corrupted video IDs, and mapped category IDs to category names.
- **Text Preprocessing:** Cleaned titles and tags while preserving multilingual text. Title and tag text were combined for sentiment analysis.
- **Sentiment Analysis:** Applied the CardiffNLP XLM-RoBERTa multilingual sentiment model to classify records as negative, neutral, or positive and retained the model confidence score.
- **SQL Analysis:** Created an SQLite analytical table and queried category performance, country-category comparisons, trending duration, monthly unique trending videos, and sentiment versus average views.
- **Power BI Dashboard:** Built four pages Executive Overview, Category Analysis, Regional Analysis, and Sentiment & Trends with KPI cards, category and country comparisons, trending-duration analysis, and sentiment time series.

5. Key Findings

- **Category Performance:** Music had the highest average views among categories with substantial record counts. Very small categories such as Trailers and Movies were treated cautiously because their sample sizes were limited.
- **Regional Performance:** Great Britain recorded the highest total views and the highest average views per trending record in this combined dataset.
- **Trending Duration:** Among videos appearing at least twice in a country, Great Britain had the highest average tracked trending duration at 12.66 days, followed by the United States at 7.12 days.
- **Sentiment:** Neutral sentiment dominated all five countries. Overall, 168,316 records were classified as neutral, 17,232 as negative, and 8,429 as positive.
- **Sentiment and Views:** Neutral records had the highest average views in each country in the analysis. This represents an observed association in the dataset and should not be interpreted as a causal relationship.

6. Conclusion

The project demonstrates an end to end data analytics workflow, from raw multicountry data preparation to NLP, SQL analysis, visualization, and data storytelling. The final dashboard makes it possible to compare content categories and countries, examine engagement patterns, understand how long videos remain in trending datasets, and explore sentiment patterns over time. The analysis also shows the importance of considering sample size and distinguishing association from causation when interpreting dashboard results.